CreditNirvana Product & Research Problem Statements

Domain: AI-native debt collections and recovery for banks, NBFCs, ARCs and fintechs in India

Common Context

CreditNirvana operates a Collections platform across a large lender base, spanning secured and unsecured retail, MSME and microfinance portfolios. A single lender portfolio can run into millions of accounts distributed across delinquency buckets (X, 1–30, 31–60, 61–90, 90+ / NPA), served by a channel stack whose fully-loaded cost per touch varies by two orders of magnitude — SMS and WhatsApp at a fraction of a rupee, AI voice bots at ₹1–3 per call, human tele-callers at ₹15–30 per connect, and field visits at ₹150–400 per productive visit.

Three constraints frame every problem below:

- **Regulatory.** RBI Fair Practices Code governs contact hours, frequency, tone and conduct. The DPDP Act governs consent, purpose limitation and data retention. Every model output must be explainable and auditable.
- **Counterfactual data.** Historical data reflects the decisions of an incumbent rule-based policy. Outcomes are only observed for actions that were actually taken. Naive supervised learning on this data will reproduce and amplify existing bias.
- **Cost asymmetry.** The objective is never "maximum recovery." It is recovery net of cost, net of customer goodwill, net of compliance risk.

Each problem statement below is one page. Data availability, sandbox access, domain SMEs and annotated ground truth can be provided by CN for all three.

Problem Statement 1: Real-Time Fake PTP Detection

Background

A promise to pay (PTP) is the most common positive outcome of a collections call, and also one of the least reliable. More than half of captured PTPs never turn into a payment.

Each broken PTP is costly in three ways. The account is parked until the promised date, so days of possible contact are lost. When the promise breaks, the account comes back through a more expensive channel. And because agents and bots are measured partly on PTP counts, inflated numbers hide what is actually being recovered.

Broken PTPs don't all break for the same reason, and the right response depends on the kind of promise:

Type	What's going on	Right response
Genuine and feasible	Borrower means it and can pay	Park until the due date, remind a day before
Genuine but not feasible	Borrower means it, but the amount or date doesn't fit their cash flow	Agree a date after salary, or a smaller amount; consider restructuring
Escape promise	Borrower says yes to end the call	Ask for a part payment now, follow up sooner
Third-party promise	Spouse, parent or employee promises on the borrower's behalf	Confirm with the borrower before parking the account
Repeat promiser	Long history of broken PTPs	Move to a stronger treatment with a shorter parking window
Agent-recorded or agent-pushed	Agent logs a PTP the borrower never really gave, or forces a date to close the call	Send to integrity review, call the borrower to confirm

Objective

A model that estimates, during the call itself, how likely the PTP under discussion is to be kept, and which of the types above it resembles. While the borrower is still on the line, the agent or bot receives a prompt to firm up a weak promise, for example by asking for ₹2,000 now through a payment link, or by moving the date to after salary credit. After the call, follow-up matches the credibility of the promise, instead of every account getting the

same fixed parking period.

Key things to solve:

- Deciding before the call ends. The model has to work on a live, partial, code-mixed transcript and audio stream, update within a few hundred milliseconds, and know when it has heard enough to act.
- Defining "kept". Late, partial and other-channel payments all blur the label, so the definition has to be agreed with the business before training starts.
- Telling insincere from infeasible. Treating genuine hardship as an escape promise puts pressure on a vulnerable borrower, so the model has to indicate why a promise may break, and that mistake has to be weighted heavily.
- Measuring real value. Prompts change the outcome the model is predicting, so accuracy on old calls isn't enough and a randomised trial is needed.
- Spotting agent-recorded PTPs. In payment data these look the same as insincere promises; only the conversation itself (did the borrower ever say a date or amount?) separates them.
- Staying hard to game. Agents, and any voice bot trained on outcomes, will adapt to the scorer, so it must stay independent of them and be monitored for drift.
- Calibrating per portfolio. A score of 0.6 should mean the same thing for a salaried personal-loan borrower and a microfinance borrower.

Key signals

- What was said: hedging ("dekhta hoon", "try karunga"), conditions ("salary aayi toh"), vague dates ("next week", "month end"), and whether the borrower stated an amount or only the agent did.
- How the conversation went: who proposed the date, whether there was any negotiation, how quickly the borrower agreed, and whether the promise came right after pressure or right before hanging up.
- How it was said: hesitation and stress in the voice. Useful as one input, never on its own.
- Account history: past PTP keep rate, whether the promised date lines up with known salary credits, bounce reasons, and an ability-to-pay estimate where available.
- What happens after the call: whether the borrower opens the payment link, replies to the confirmation message or calls

back.
- The agent: their PTP-to-payment rate compared with peers working similar accounts, PTPs bunched at the end of a shift, and remarks copied across accounts.

Expected outcome

A PTP credibility service running inside the CN platform, integrated as follows:

- Input from CN: live call audio and transcripts from the CN voice bot and tele-calling stack, account and payment history, and payment-link events.
- CN agent desktop: the live credibility score, likely PTP type and in-call prompts shown to tele-callers while the borrower is on the line.
- CN voice bot: the same score and prompts used by the bot's dialogue flow to firm up weak promises.
- CN campaign and disposition engine: PTP quality sets the parking window, reminder schedule and third-party confirmation call, replacing the fixed rule.
- CN dashboards: a quality-weighted PTP measure reported alongside raw PTP counts for agents, bots and teams, plus a per-agent PTP quality view for the integrity team.

Key metrics

- Keep rate and amount recovered on calls with prompts versus calls without, in a randomised trial
- AUC and calibration of the keep prediction at the end of the call
- Share of broken PTPs flagged before the account is parked
- Rate at which hardship cases are misclassified

Constraints

A low score must never lead to harsher treatment where there are signs of hardship; those cases go to restructuring or to a human. Prompts have to stay within the Fair Practices Code, so firming up a promise must not turn into pressure. Using voice to judge credibility counts as sensitive processing under DPDP, which means purpose and retention limits apply. Each update has to fit the live-call latency budget, and every flag has to be explainable to the agent, their supervisor and an auditor.

Problem Statement 2: Right-Party Contact Prediction and Skip-Trace Prioritisation (Tele and Field)

Background

Every tele-call and field visit assumes that the phone number or address on file still reaches the borrower. In practice, a large share of dial attempts and visits go to contact points that are dead, recycled to a new subscriber, belong to a relative or former employer, or were wrong from the start. This is usually discovered slowly: a number rings out for weeks before anyone acts, or an address is visited three times before it is marked untraceable.

Skip-tracing is expensive and is normally triggered by a fixed attempt count rather than by evidence. As a result, it comes too late for the accounts that needed it and is wasted on accounts where the borrower was simply not answering.

The question that matters, whether a specific contact point still reaches a specific borrower, is rarely modelled on its own. It is usually folded into an account-level "contactability" score, which hides the actual decision: which number to dial, which address to visit, and when to stop and trace.

Contact point	State	Right response
Phone	Valid and reachable	Keep dialling, at the best time slot
Phone	Valid, but borrower avoiding	Switch channel (WhatsApp, field) instead of redialling
Phone	Temporarily unreachable	Retry later, with backoff
Phone	Switched off long-term	Move to another number, or trace
Phone	Recycled to a new subscriber	Stop at once; there is a risk of disclosing the debt to a stranger
Phone	Third party (relative, employer, reference)	Use only within Fair Practices Code rules, never discuss the debt
Phone	Invalid from the start	Trace
Address	Valid and occupied	Visit
Address	Valid, but borrower usually absent	Change the visit time
Address	Borrower has moved	Trace the new address
Address	Hard to find on the ground	Resolve the location (Problem Statement 3) before writing it off
Address	Fabricated or incomplete at origination	Trace, and flag to the origination team

Objective

A model that predicts, for every phone number and address on file, how likely it is to reach the borrower, that is, to produce a right-party contact (RPC) in tele or field. A prioritisation layer then decides, for each account, whether to keep trying, switch to

another contact point or channel, or trigger a skip-trace, with skip-traces ranked by the expected recovery value of finding the borrower.

This is a prediction made before the call or visit. Checking during a call whether the person who answered is actually the borrower is a separate identity-verification problem and is out of scope.

Key things to solve:

- Building labels. True health is never observed directly. A right-party contact is only recorded when someone dialled, the borrower answered and the agent logged it correctly, and many contact points have never been tested at all.
- Avoiding versus invalid. A borrower who won't answer and a dead number look similar on the surface but need opposite actions: switch channel for the first, trace for the second. Separating them is the central modelling problem.
- Recycled numbers. Telecom operators reassign inactive numbers to new subscribers without any explicit signal. Calling a recycled number wastes effort and risks disclosing the debt to a stranger, which breaches the Fair Practices Code.
- Health changes over time. The model needs to estimate how long a contact point is likely to stay valid, so that confidence falls as evidence ages. A survival or hidden-state model fits this better than a one-off classifier.
- When to trace. A skip-trace is worth it when the expected recovery from finding a valid contact point, given the amount outstanding and the chance of recovery once reached, is higher than the trace cost. This is a value-of-information decision, not a fixed attempt count.
- Feedback loops. Once the score decides which numbers get dialled, it also shapes the data it learns from. A small, explicit exploration budget keeps low-scored contact points from being starved of the evidence that would correct them.

Key signals

- Telephony: attempt history by time slot, ring duration, answer and immediate-hangup patterns, network responses ("switched off", "not reachable", "does not exist"), and time since the last right-party contact on the number.
- Dispositions and remarks: wrong-number and third-party dispositions, agent remarks ("phone uthaya bhai ne",

"number band hai 2 mahine se") and voice-bot transcripts showing who answered.
- Shared contact points: the same number or address appearing across several borrowers, lenders or co-borrower records, which often points to a reference, shared or recycled contact.
- Field visits: outcomes at the address, GPS dwell at check-in, and the difference between "locked premises" and "nobody of that name here".
- Record history: where the contact point came from (KYC at origination, a later update, the bureau, the borrower during a call), how old it is, and whether it was ever confirmed by a contact that led to a payment.

Suggested phasing

The core question and the final decision are the same for both channels, so this stays one problem. The data is not: phone data is dense, with many attempts a day, while field data is sparse and costly, with a few visits a month. The work is therefore phased:

- Weeks 1 to 6: phone numbers. Build health scoring for phone contact points, the avoiding-versus-invalid split and the recycled-number check, and integrate with the CN dialer and the compliance suppression list.
- Weeks 7 to 12: addresses and field evidence. Add address health, bring field visit outcomes into the same model, use location confidence from Problem Statement 3, and switch on the account-level skip-trace decision across all contact points.

Expected outcome

A right-party contact prediction service running inside the CN platform, integrated as follows:

- Input from CN: dial attempt logs and network responses from the CN dialer, dispositions and remarks, voice-bot transcripts, field visit outcomes from the CN field app, and contact-point source and history from account records.
- CN dialer: dial lists ordered by contact-point health, so the best number for each account is tried first and dead numbers drop out.
- CN campaign engine: the recommended action for each account (continue, switch contact point, switch channel, trace), with a reason code, applied as campaign rules.
- CN skip-trace queue: accounts ranked by the expected value

of tracing, replacing the fixed attempt-count trigger.
- CN compliance controls: a suppression list for recycled and third-party numbers, so the debt is never discussed with the wrong person.
- CN visit planner: address health used in visit selection, alongside location confidence from Problem Statement 3.

Key metrics

- Right-party contact rate per thousand dial attempts and per hundred field visits, compared with current rules
- AUC and calibration of the right-party contact prediction for each contact point
- Attempts and visits spent on invalid contact points before action is taken
- Skip-trace hit rate and recovery from traced accounts, compared with the fixed attempt-count rule
- Third-party disclosure incidents (target: zero)

Constraints

The debt is never disclosed to a third party, as required by the Fair Practices Code. Contact data from bureaus and references is used only for the purpose it was collected for, under DPDP. Skip-trace sources must be lawful and auditable. Scores are refreshed daily so that they are ready for outbound planning.

Problem Statement 3: Address Geocoder That Learns from Field Visits

Background

Field collections depends on finding the borrower, and that depends on an address that is often a description rather than a location: "near Hanuman temple, behind ration shop, 2nd cross, Kamakshipalya". Commercial geocoders place addresses like this at the pincode or locality centre, which in tier-2, tier-3 and rural areas can be hundreds of metres or even several kilometres from the actual house. Agents spend much of their day searching and asking neighbours. Eventually they mark accounts "address not traceable", including some where the borrower lives exactly where the address says.

Meanwhile, the answer already exists for a large share of accounts: the GPS location where an agent actually met the borrower on an earlier visit. Today that sits in visit logs and never flows back into the address record.

This problem is about where an address is. Whether the borrower still lives there is covered by Problem Statement 2. The two are linked: the confidence score from this work helps Problem Statement 2 tell "couldn't find it" apart from "doesn't exist".

Objective

A geocoder that gives every account address a location pin, a confidence radius and simple landmark-based directions for the agent. It learns from where field agents actually found borrowers and keeps improving as new visits come in.

Key things to solve:

- Weak, noisy evidence. GPS on a successful visit is a clue, not ground truth, since the borrower may have been met at work or on the road. Each piece of evidence has to be weighted by reliability and combined into one estimate with an uncertainty attached.
- Reading Indian addresses. Mixed scripts, transliterations, abbreviations and relative landmarks ("Hanuman Mandir ke peeche" vs. "behind Hanuman temple") need multilingual entity extraction and landmark resolution, not string matching.
- One place, many spellings. Resolving the same building across accounts and lenders into a single location, so that one confirmed visit improves many addresses.
- Honest confidence. The visit planner uses it to decide whether to visit or verify first, and an uncalibrated score is worse than none.
- Fake visits. Check-ins from an agent's home or a nearby tea stall would teach the geocoder wrong locations, so low-integrity evidence gets less weight and no single agent can push the model off course.

Available evidence

- The address text itself: landmarks, cross roads, building names and localities, written in mixed scripts, transliterated and often misspelt.
- GPS from successful visits, which shows where the borrower was met. They may have been met at their shop or on the street, and GPS accuracy varies.
- GPS from failed visits, which shows where the agent searched.
- Dwell time and the shape of the GPS trail. Searching looks

different from walking straight to a known house, although this depends on how well the agent knows the area.
- Agent remarks, which sometimes contain corrections ("actual house 2 gali aage").
- Confirmed locations of other nearby accounts that share the same landmarks.

Expected outcome

A geocoding service running inside the CN platform, integrated as follows:

- Input from CN: account addresses, visit logs with GPS trails and dwell times, visit outcomes and agent remarks from the CN field app, and visit-integrity signals.
- CN field app: the corrected pin, confidence radius and landmark directions shown for each visit, available offline.
- CN visit planner: pins and confidence used for visit selection and routing, with low-confidence addresses sent for verification before a visit is planned.
- CN address records: confirmed locations written back to the account and shared across accounts at the same place, so each successful visit improves the record.
- Right-party contact prediction (Problem Statement 2): location confidence passed in, so addresses that are only hard to find are not marked invalid.
- CN dashboards: geocoding quality and "not traceable" trends by territory for field managers.

Key metrics

- Median and 90th-percentile distance error against confirmed visit locations, compared with a commercial geocoder
- Share of "address not traceable" outcomes in a pilot territory
- Productive visits per agent-day in a controlled comparison
- Calibration of the confidence radius

Constraints

Location data is used only for collections, within DPDP retention limits. Access is restricted so that field operations data stays separate from other uses. The system does not infer locations beyond what collections needs. Agent GPS is used only within the applicable employment-law and agent-monitoring framework. Output has to work offline, because many field areas have poor connectivity.